CISCO FMC / FTD

Basic Deployment & Internet Access Configuration Guide

A complete lab walkthrough covering FMC/FTD initial setup, device registration, interface configuration, DHCP services, static routing, NAT policy, and access control — from first boot to verified internet connectivity.

Author: **Abdelrhman Nabil**
Role: OT/ICS Security Engineer

Category: Network Security Lab · Portfolio Project
Platform: Cisco Firepower (FMC v6.7 / FTD v6.4)

Table of Contents

1. Introduction

This guide provides a step-by-step walkthrough for deploying Cisco Firepower Management Center (FMC) and Firepower Threat Defense (FTD) in a lab environment. The objective is to establish centralized management of the FTD appliance through FMC and configure basic internet access for internal hosts.

By the end of this guide, you will have a fully functional FMC-managed FTD deployment with inside-to-outside internet connectivity verified through ping tests to public DNS servers.

1.1 Scope & Objectives

- Deploy FMC and FTD virtual appliances in a controlled lab environment
- Register FTD with FMC for centralized management
- Configure network interfaces with appropriate security zones
- Enable DHCP for automatic host addressing on the inside network
- Establish internet connectivity through static routing, NAT, and access control policies

2. Lab Environment

2.1 Network Topology

The lab topology consists of the FMC management server, an FTD security appliance acting as the perimeter firewall, two Windows PCs (one for management, one as an inside host), a layer-2 switch for management connectivity, and a cloud network simulating the internet.

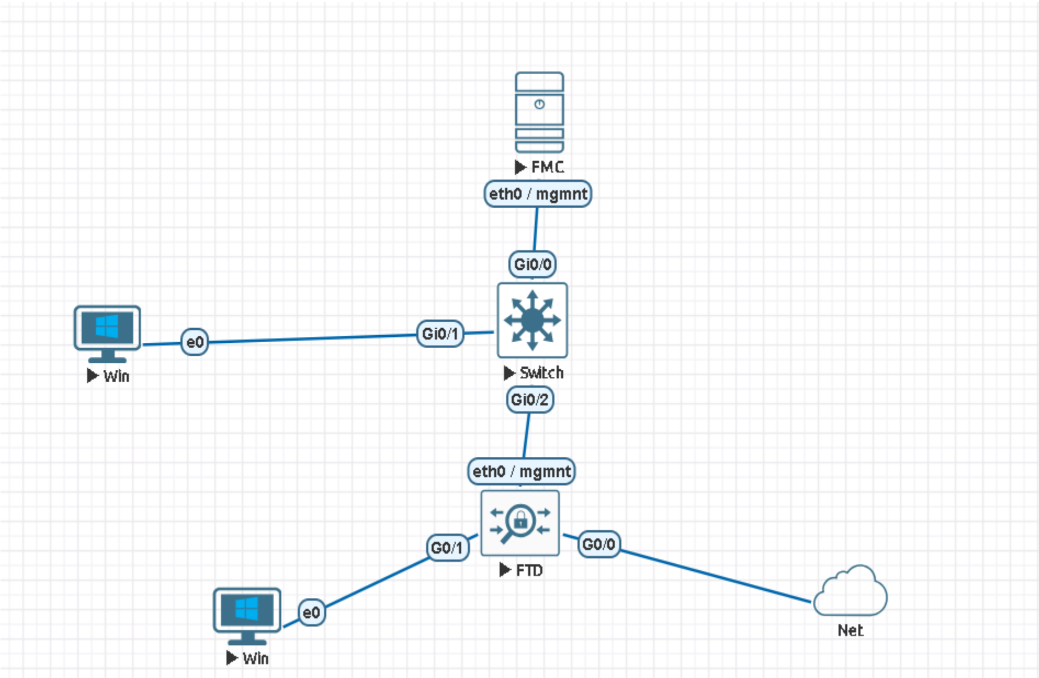

Figure 1 — Lab Network Topology

2.2 Software Versions

- **FMC (Firepower Management Center):** v6.7
- **FTD (Firepower Threat Defense):** v6.4
- **Windows PCs:** Windows 7
- **Cloud:** Simulates outside network (internet access)
- **Switch:** Layer-2 management connectivity between all devices

2.3 System Resources

- **FMC:** 16 GB RAM
- **FTD:** 8 GB RAM
- **PCs & Switch:** 1 GB RAM each

2.4 IP Addressing Scheme

The management network uses the 192.168.100.0/24 subnet. All devices are assigned static addresses for initial connectivity:

Device	Role	IP Address
Management PC	FMC GUI Access	192.168.100.100
FMC	Management Server	192.168.100.10
FTD	Security Appliance	192.168.100.1

3. Initial Setup & Connectivity

3.1 Connectivity Verification

Before beginning any configuration, verify Layer-3 reachability between all management interfaces using ICMP ping:

- Management PC ↔ FMC
- FMC ↔ FTD
- Management PC ↔ FTD

All three pairs must respond successfully before proceeding to device registration.

3.2 Accessing the FMC GUI

1. From the Management PC, open a web browser.
2. Navigate to https://192.168.100.10 (FMC management IP).
3. Log in with the default credentials:
    - **Username:** admin
    - **Password:** Admin123
4. Change the default password when prompted by the setup wizard.
5. Activate the 90-day evaluation license to unlock all features.

4. Registering FTD with FMC

The FTD must be registered with the FMC to enable centralized policy management. During initial FTD setup, ensure you select FMC as the management mode — not local management. Communication between FTD and FMC uses TCP port 8305 (outbound from FTD to FMC).

4.1 FTD CLI — Configure Manager

On the FTD command-line interface, run the following command to point the appliance at the FMC:

```
> configure manager add 192.168.100.10 key123
```

- 192.168.100.10 → FMC management IP address
- key123 → Shared registration key (must match FMC-side configuration)

Figure 2 — FTD CLI: configure manager command

4.2 Verify Registration Status

Run the show managers command to confirm the FMC registration is in a pending state:

```
> show managers
```

The FMC should appear with a "Pending" status. This is expected — the registration will complete once confirmed from the FMC GUI.

Figure 3 — show managers output (Pending status)

4.3 Confirm Registration from FMC

Complete the registration from the FMC web interface:

1. Log in to the FMC GUI.
2. Navigate to Devices → Device Management → Add Device.
3. Enter the registration details:
   - **Host:** 192.168.100.1 (FTD management IP)
   - **Registration Key:** key123 (must match the FTD-side key)
   - **Display Name:** Choose a meaningful name (e.g., FTD-Lab)
   - **Access Control Policy:** Select an existing policy or create one (e.g., Lab-Access-Policy)
4. Click Register to initiate the secure connection.
5. Once successful, the FTD status changes from Pending to Online.

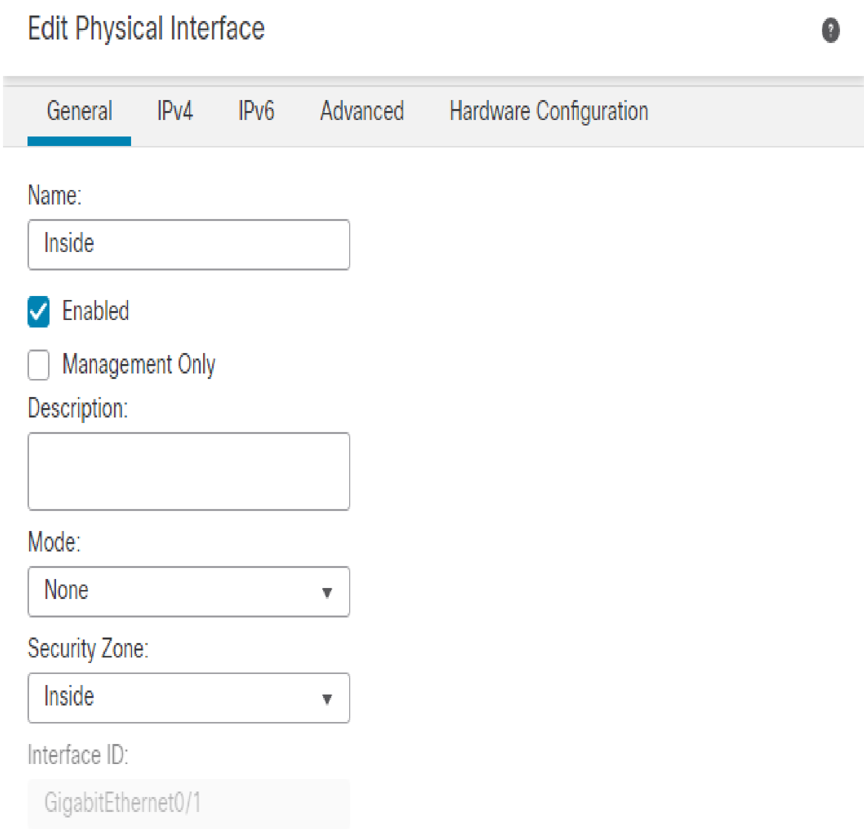

Figure 4 — FMC Add Device dialog

Registration Confirmation

After successful registration, the FMC shows "Communication with FTD has been established, discovery in progress" and the FTD CLI confirms Registration: Completed.

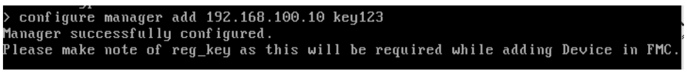

Figure 5 — FMC registration confirmation

5. Post-Registration Configuration

With the FTD now under FMC management, all further configuration is performed through the FMC web interface. The deployment button must be clicked after each configuration change to push policies to the FTD.

5.1 Configure FTD Interfaces

Navigate to Devices → Device Management → select your FTD → Edit → Interfaces tab. Configure each physical interface with a logical name, security zone, and IP address:

1. Select a physical interface (e.g., GigabitEthernet0/0).
2. Assign a logical name (e.g., Outside or Inside).
3. Create or assign a Security Zone (e.g., OUTSIDE or INSIDE).
4. Configure the IPv4 address — choose Static or DHCP and enter the IP/mask.
5. Optionally enable Management Access if you need to manage or ping via this interface.
6. Click Save, then Deploy the changes to push configuration to the FTD.

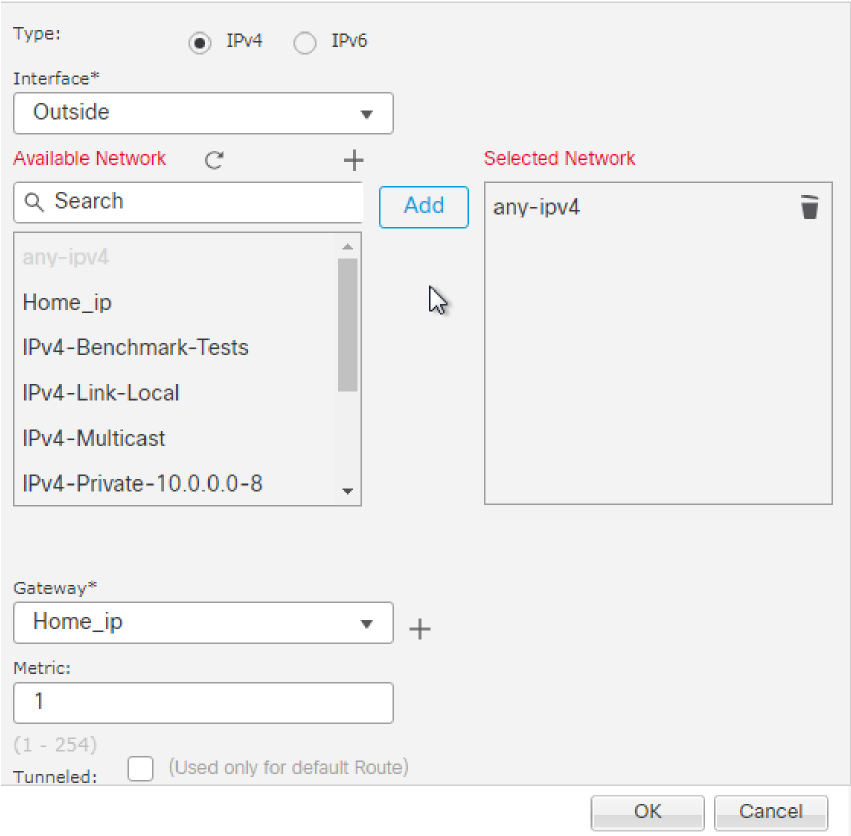

Figure 6 — Edit Physical Interface dialog

Configured Interface Summary

Interface	Security Zone	IP Address
GigabitEthernet0/0	**Outside**	192.168.1.11/24 (Static)
GigabitEthernet0/1	**Inside**	192.168.2.1/24 (Static)

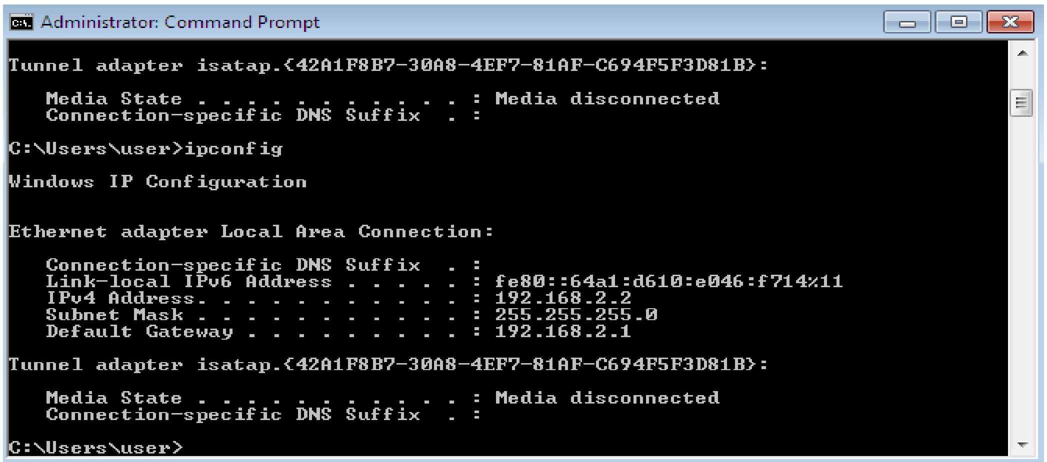

Figure 7 — FMC Interfaces tab after configuration

5.2 Configure DHCP Server for Inside Zone

To automatically assign IP addresses to hosts on the inside network, configure a DHCP server on the inside interface:

1. In FMC, go to Devices → Device Management → select your FTD → Edit.
2. Navigate to the DHCP tab.

3. Under DHCP Server Configuration, click Add.
4. Select the Inside interface and define the address pool (e.g., 192.168.2.2 – 192.168.2.254).
5. Deploy the changes.

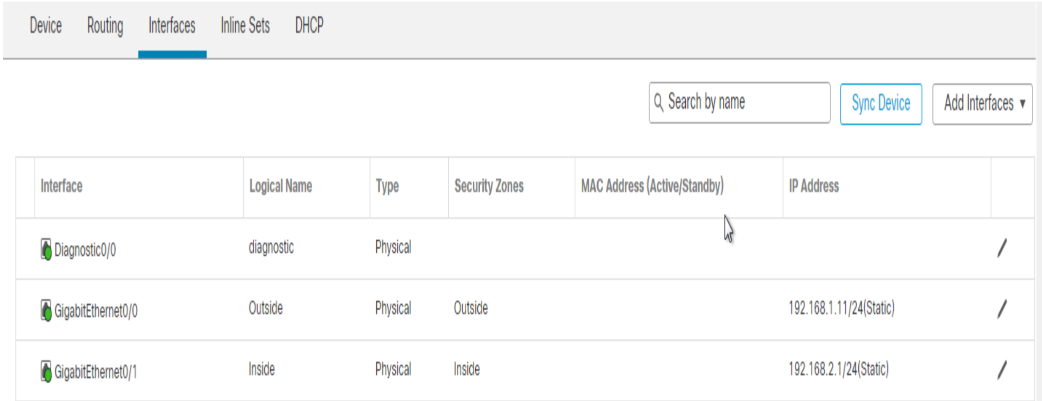

Figure 8 — DHCP configuration and client verification (ipconfig)

6. Internet Access Configuration

Enabling internet access for inside hosts requires three components working together: a default static route pointing outbound traffic to the upstream gateway, a NAT policy translating private inside addresses to the outside interface IP, and an access control rule permitting the traffic flow.

6.1 Static Route

Configure a default route pointing all unknown-destination traffic out the outside interface toward the upstream gateway:

1. Navigate to Devices → Device Management → FTD → Routing → Static Route.
2. Click Add Route and configure:
   - **Interface:** Outside
   - **Network:** 0.0.0.0/0 (default route — matches all destinations)
   - **Gateway:** Next-hop IP of the upstream cloud/internet network

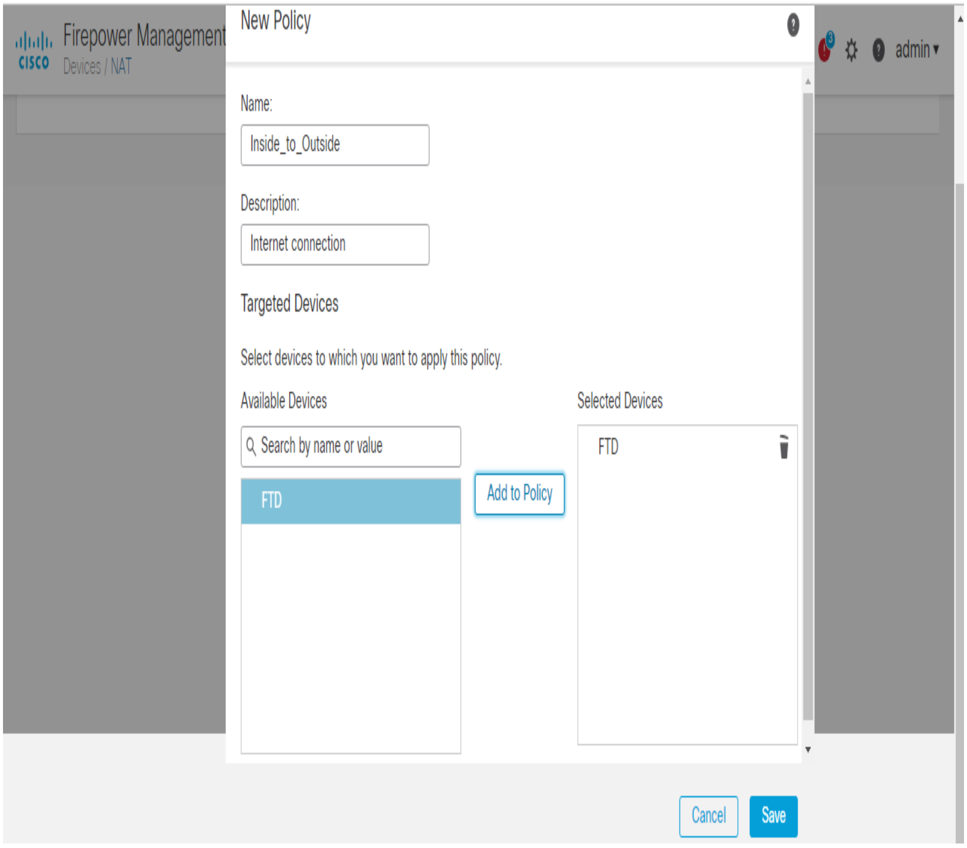

Figure 9 — Static Route configuration

6.2 NAT Policy

Create a NAT policy to translate inside source addresses to the outside interface IP (PAT / dynamic NAT overload):

1. Navigate to Devices → NAT.
2. Create a New Policy → select Threat Defense NAT → assign to the FTD device.
3. Add a Manual NAT Rule with the following settings:
   - **Type:** Dynamic
   - **Source Interface:** Inside
   - **Destination Interface:** Outside
   - **Original Source:** IPv4 inside network (e.g., 192.168.0.0/16)
   - **Translated Source:** Destination Interface IP
4. Save and Deploy.

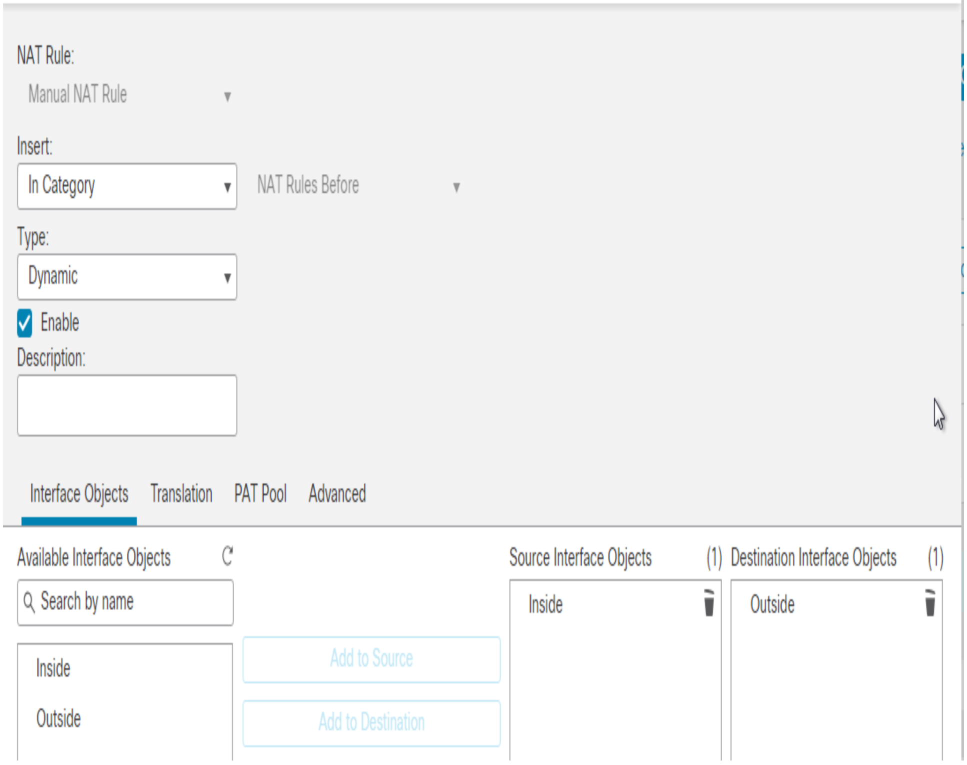

Figure 10 — NAT policy creation (assign to FTD)

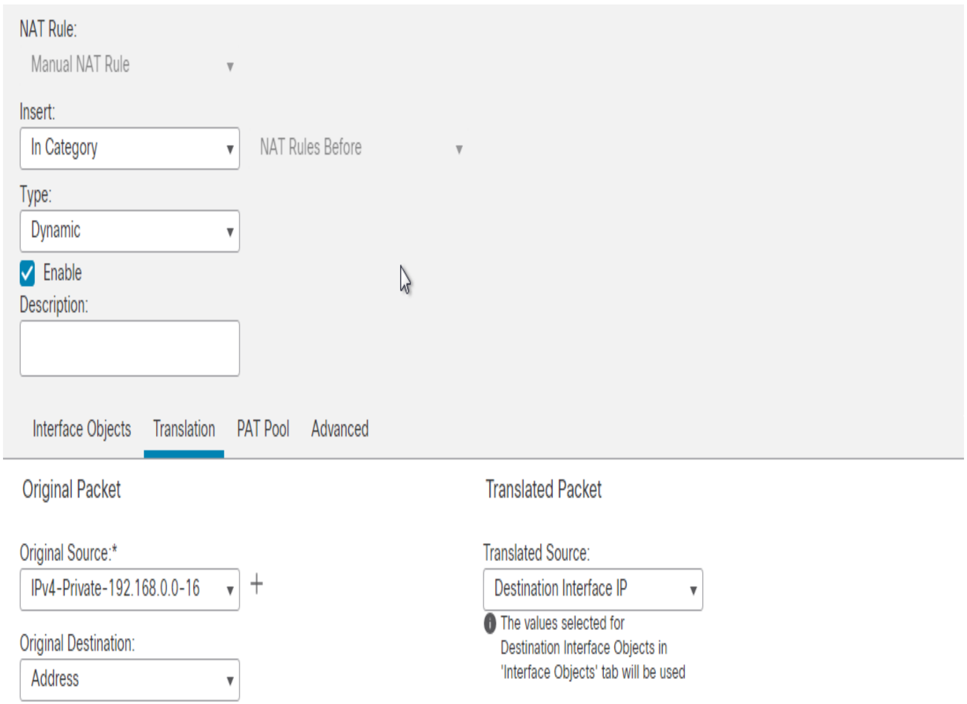

Figure 11 — NAT rule: Interface Objects and Translation tabs

6.3 Access Control Policy

The final component is an Access Control rule that allows traffic from the Inside zone to the Outside zone:

1. Navigate to Policies → Access Control → select your policy.
2. Add a new rule:
   - **Name:** Internet Access
   - **Action:** Allow

   - **Source Zone:** Inside
   - **Destination Zone:** Outside

3. Save and Deploy the policy.

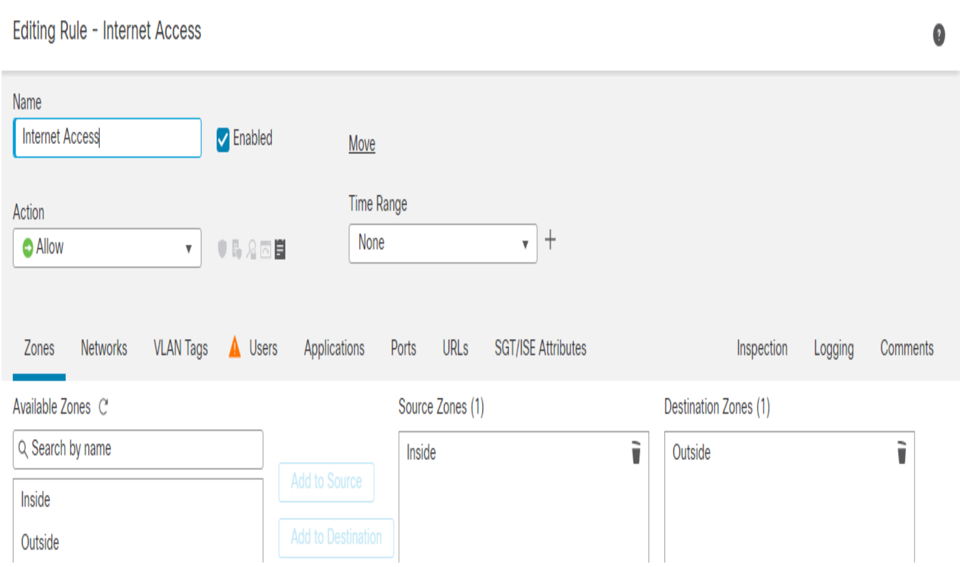

Figure 12 — Access Control rule: Inside → Outside (Allow)

7. Verification & Results

After deploying all three components (static route, NAT, and access control), verify internet connectivity from an inside host by pinging a public IP address such as Google DNS (8.8.8.8):

```
C:\Users\user> ping 8.8.8.8
```

The first attempt may time out (100% loss) due to deployment propagation delay. A second attempt confirms successful connectivity with 0% packet loss and response times averaging ~83ms.

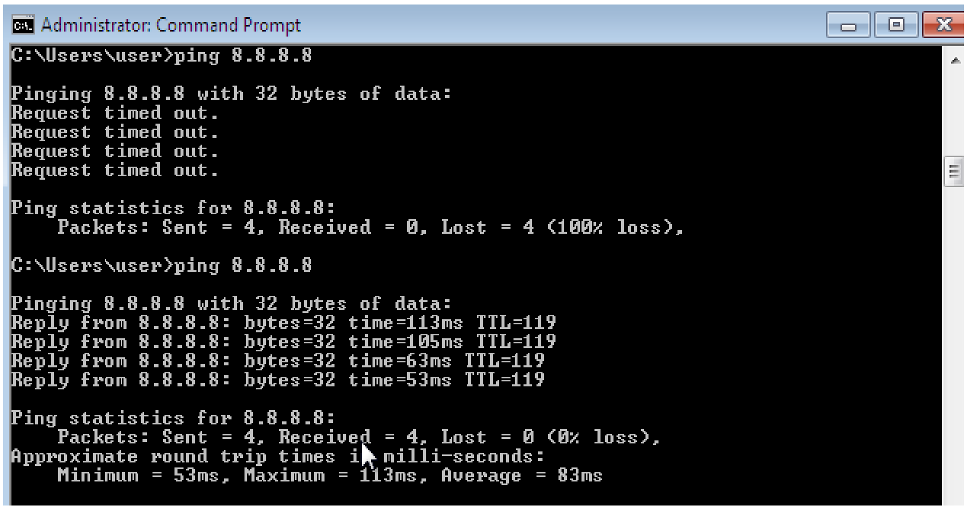

Figure 13 — Ping verification: 8.8.8.8 reachable from inside host

This confirms the complete FMC/FTD deployment pipeline is operational — from initial device registration through to verified internet access for inside-zone hosts.

8. Summary

This lab demonstrated a complete end-to-end deployment of Cisco Firepower Management Center and Firepower Threat Defense, covering the following milestones:

- FMC and FTD virtual appliance deployment with proper resource allocation
- FTD registration with FMC using shared key authentication (TCP 8305)
- Interface configuration with security zone assignment (Inside/Outside)
- DHCP server deployment for automatic inside host addressing
- Default static route configuration for outbound traffic
- Dynamic NAT (PAT) policy enabling private-to-public address translation
- Access control policy permitting Inside-to-Outside traffic flow
- End-to-end verification via successful ping to 8.8.8.8